

MULTI-OS OFFENSIVE SECURITY LABORATORY INFRASTRUCTURE SETUP REPORT

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Document Type: Lab Architecture & Deployment Blueprint

Classification: Internal — Environment Configuration Records

SECTION 1: LABORATORY OBJECTIVE & ENGINEERING GOALS

The primary engineering objective is to build a fully functional, multi-OS penetration testing sandbox on a single physical host. The lab supports safe, ethical and controlled execution of network reconnaissance, vulnerability scanning, and exploitation workflows without exposing production assets.

Using a Type-2 hypervisor with a Bridged Network topology, the environment simulates an on-site internal network audit. It serves as a stable platform for practising offensive techniques and documenting proof-of-concept attacks.

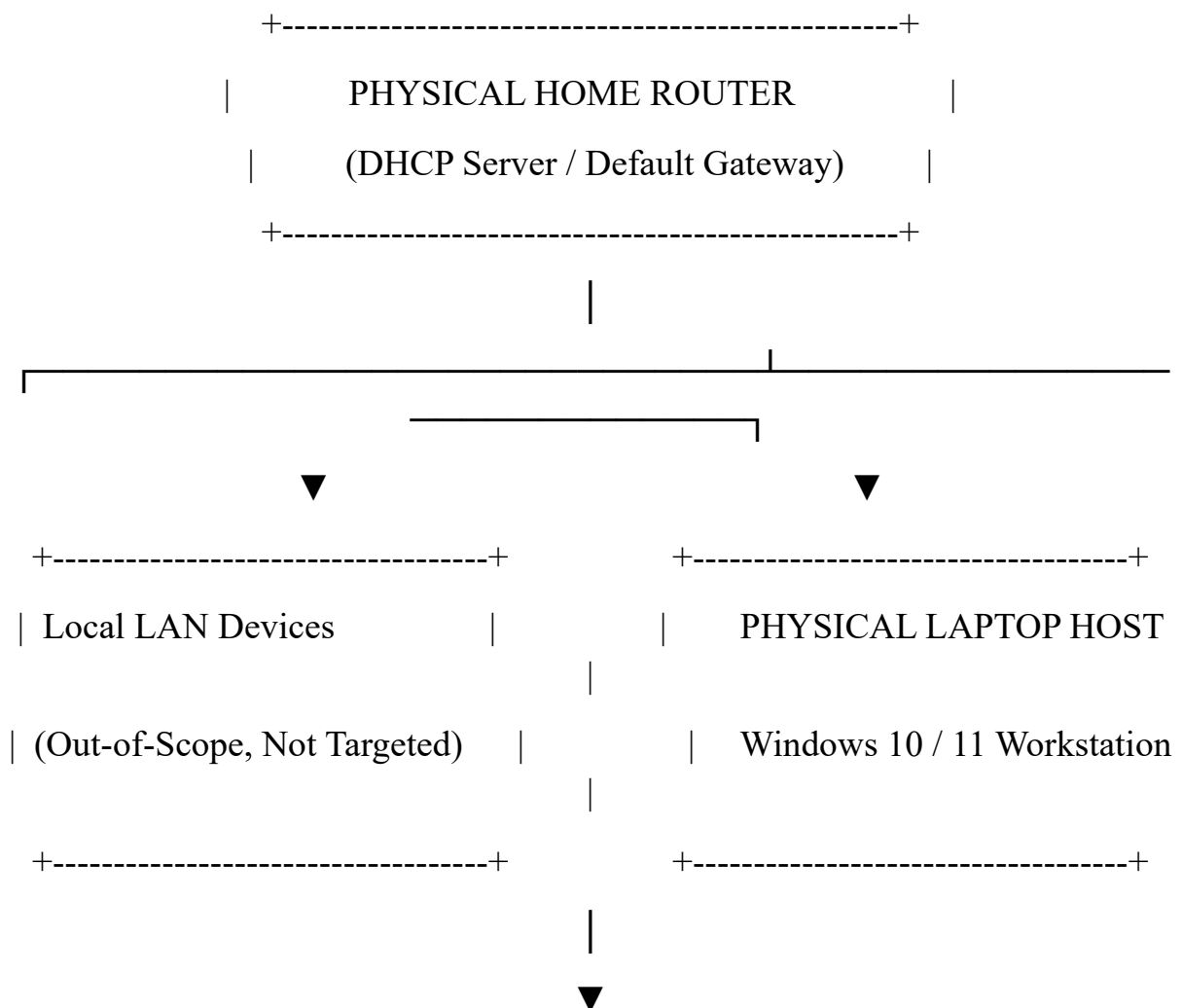
SECTION 2: HOST HARDWARE & HYPERVISOR CONFIGURATION

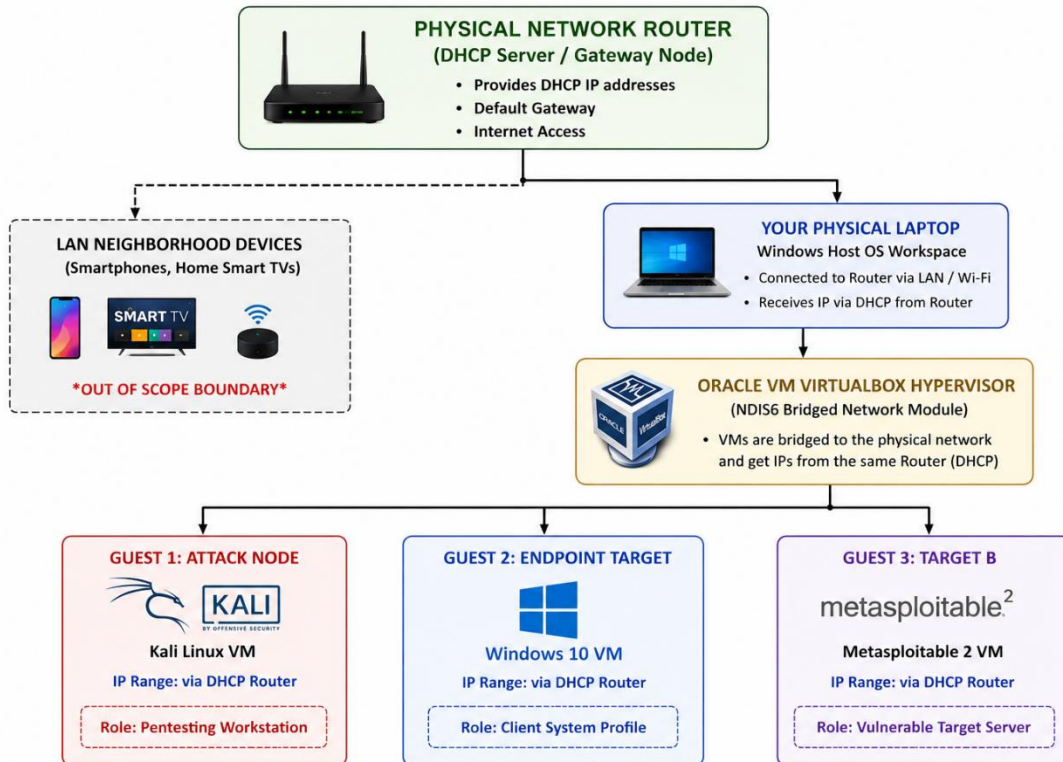
- Host OS: Microsoft Windows 10 / 11 Enterprise (64-bit)
- Hypervisor: Oracle VM VirtualBox v7.x (Type-2)
- Virtualisation Extensions: Intel VT-x / AMD-V enabled in UEFI/BIOS
- Resource Allocation:

- Host margin: minimum 4 GB RAM reserved for Windows
- Guest VMs: thin-provisioned disks and dedicated RAM slices

SECTION 3: NETWORK TOPOLOGY – BRIDGED MODE

The VirtualBox NDIS6 Bridged Networking Driver attaches directly to the host's physical adapter. Guest VMs appear as independent nodes on the local router's subnet.





Network Topology Diagram

SECTION 4: VIRTUAL MACHINE INVENTORY

4.1 Kali Linux (Attack Node)

- Debian Testing 64-bit
- 2 vCPUs, 4096 MB RAM, 40 GB disk
- IP: 192.168.1.242 (Bridged, DHCP)
- Tools: Nmap, Metasploit, Netcat, Wireshark

```
kali@kali: ~  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.1.242 netmask 255.255.255.0 broadcast 192.168.1.255  
    inet6 fe80::a88:27ff:fe9a:78e6 prefixlen 64 scopeid 0x20<link>  
    inet6 2401::4900:8902:65fc:801b:0570:7a2a:db1a prefixlen 64 scopeid 0x0<global>  
    inet6 2401::4900:8902:65fc:a00:27ff:fe9a:78e6 prefixlen 64 scopeid 0x0<global>  
    ether 08:00:27:9a:78:e6 txqueuelen 1000 (Ethernet)  
    RX packets 132 bytes 24007 (24.0 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 67 bytes 11552 (11.2 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
kali@kali: ~
```

Kali linux ifconfig

4.2 Windows 10 (Endpoint Target)

- Windows 10 Enterprise/Pro 64-bit
- 2 vCPUs, 4096 MB RAM, 50 GB disk
- IP: 192.168.56.1 (Host-Only adapter)
- Simulates corporate workstation

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::92bf:83e5:9ce7:61a7%17  
IPv4 Address. . . . . : 192.168.56.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :  
  
Wireless LAN adapter Local Area Connection* 1:  
  
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :  
  
Wireless LAN adapter Local Area Connection* 2:  
  
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :  
  
Wireless LAN adapter Wi-Fi:  
  
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2401::4900:8902:65fc:e32e:3ecf:42b5:b58b  
Temporary IPv6 Address. . . . . : 2401::4900:8902:65fc:28b9:dcd0:368f:5e3c  
Link-local IPv6 Address . . . . . : fe80::f1d5:9164:2d41:108%16  
IPv4 Address. . . . . : 192.168.1.7  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::6e4f:89ff:fe1a:b5ab%16  
192.168.1.1
```

Windows ipconfig

4.3 Metasploitable 2 (Vulnerable Server)

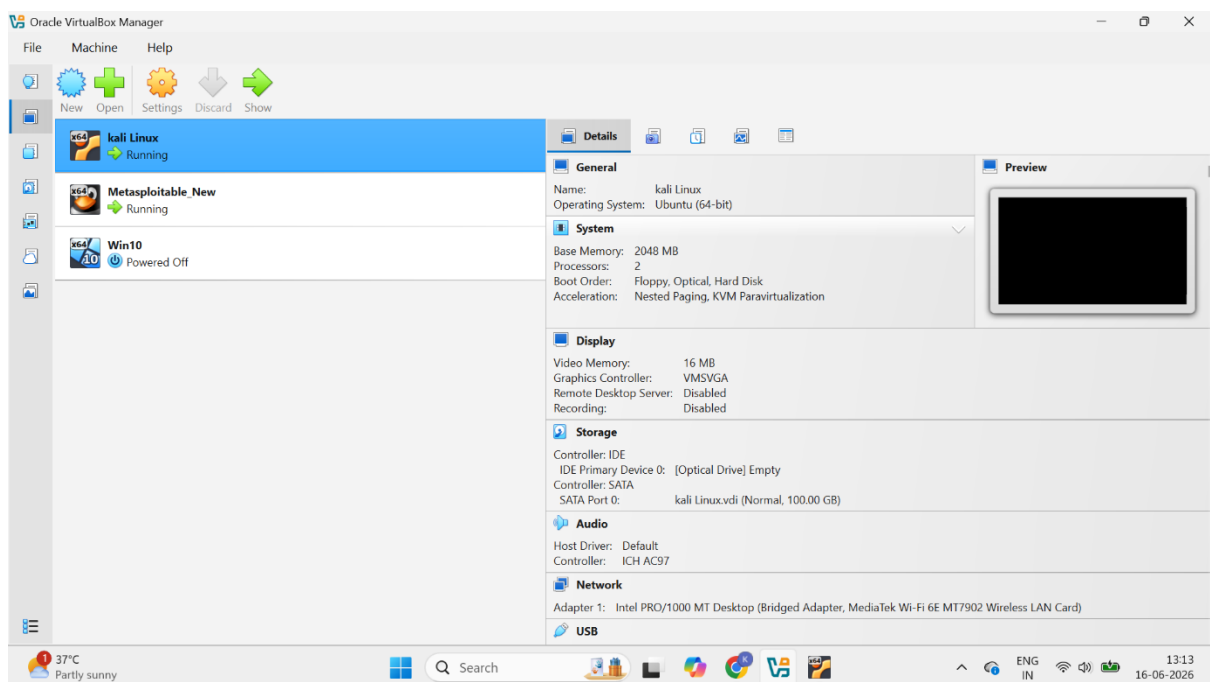
- Ubuntu 8.04 Server (intentionally unpatched)
- 1 vCPU, 1024 MB RAM, 10 GB disk
- IP: 192.168.1.21 (Bridged, DHCP)
- Over 20 open ports, legacy services

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:bd:87:44
          inet addr:192.168.1.21  Bcast:192.168.1.255  Mask:255.255.255.0
          inet6 addr: 2401:4900:8902:65fc:a00:27ff:febd:8744/64 Scope:Global
          inet6 addr: fe80::a00:27ff:febd:8744/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:61 errors:0 dropped:0 overruns:0 frame:0
          TX packets:55 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5750 (5.6 KB)  TX bytes:6130 (5.9 KB)
          Base address:0xd020 Memory:f0200000-f0220000

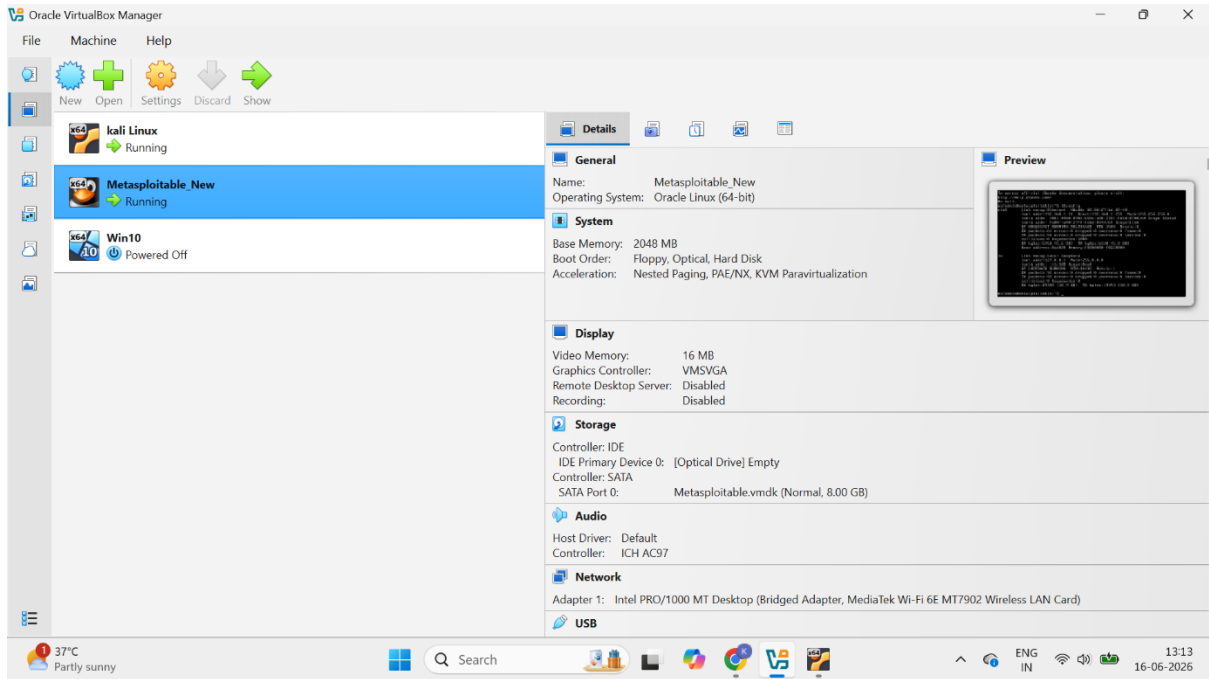
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$ _
```

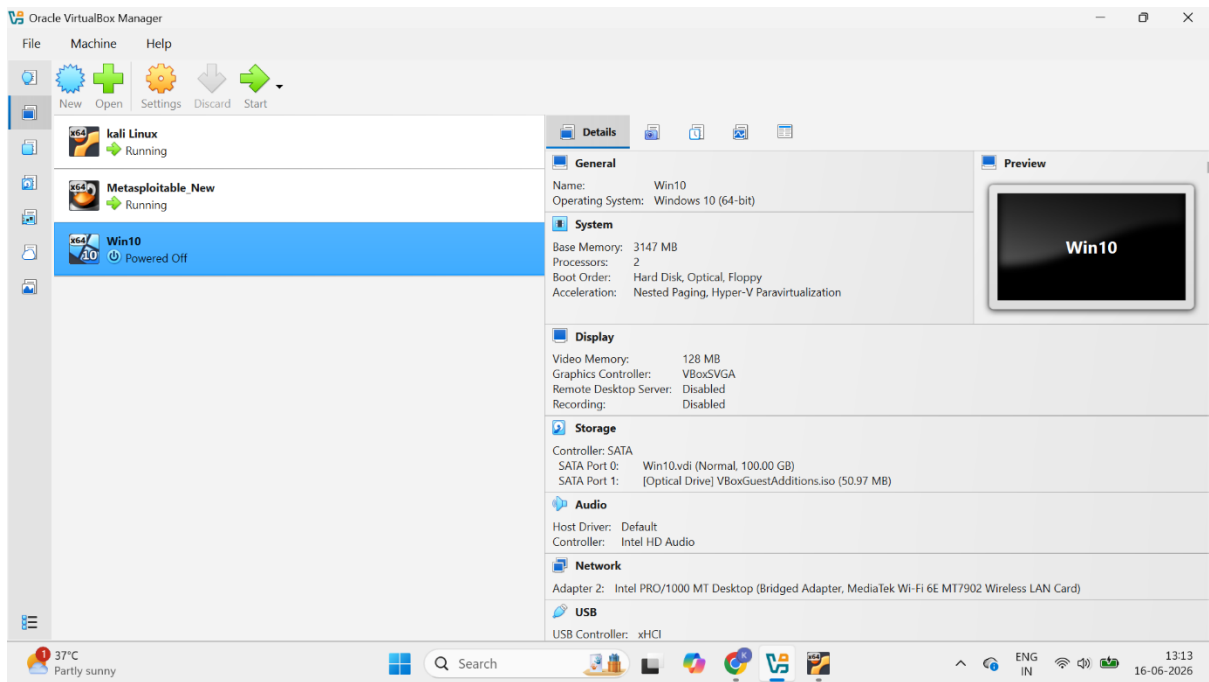
Metasploitable 2 ifconfig



VirtualBox Manager showing kali



VirtualBox Manager showing Metasploitable 2



VirtualBox Manager showing
Windows 10 endpoint

SECTION 5: OPERATIONAL READINESS CHECKS

- All VMs boot and obtain correct IP addresses
- Kali → Metasploitable ping successful
- Hypervisor snapshots taken before exploitation
- Out-of-scope devices unreachable
-

SECTION 6: INFRASTRUCTURE SUMMARY

The lab is fully deployed on a single laptop using Oracle VirtualBox with a bridged network. Three VMs are operational:

- Kali Linux attacker (192.168.1.242)
- Windows 10 endpoint (192.168.56.1)
- Metasploitable 2 target (192.168.1.21)

Warning !

Strict scoping and snapshots ensure ethical, legal and repeatable testing. The infrastructure provides a robust platform for penetration testing and skill demonstration.